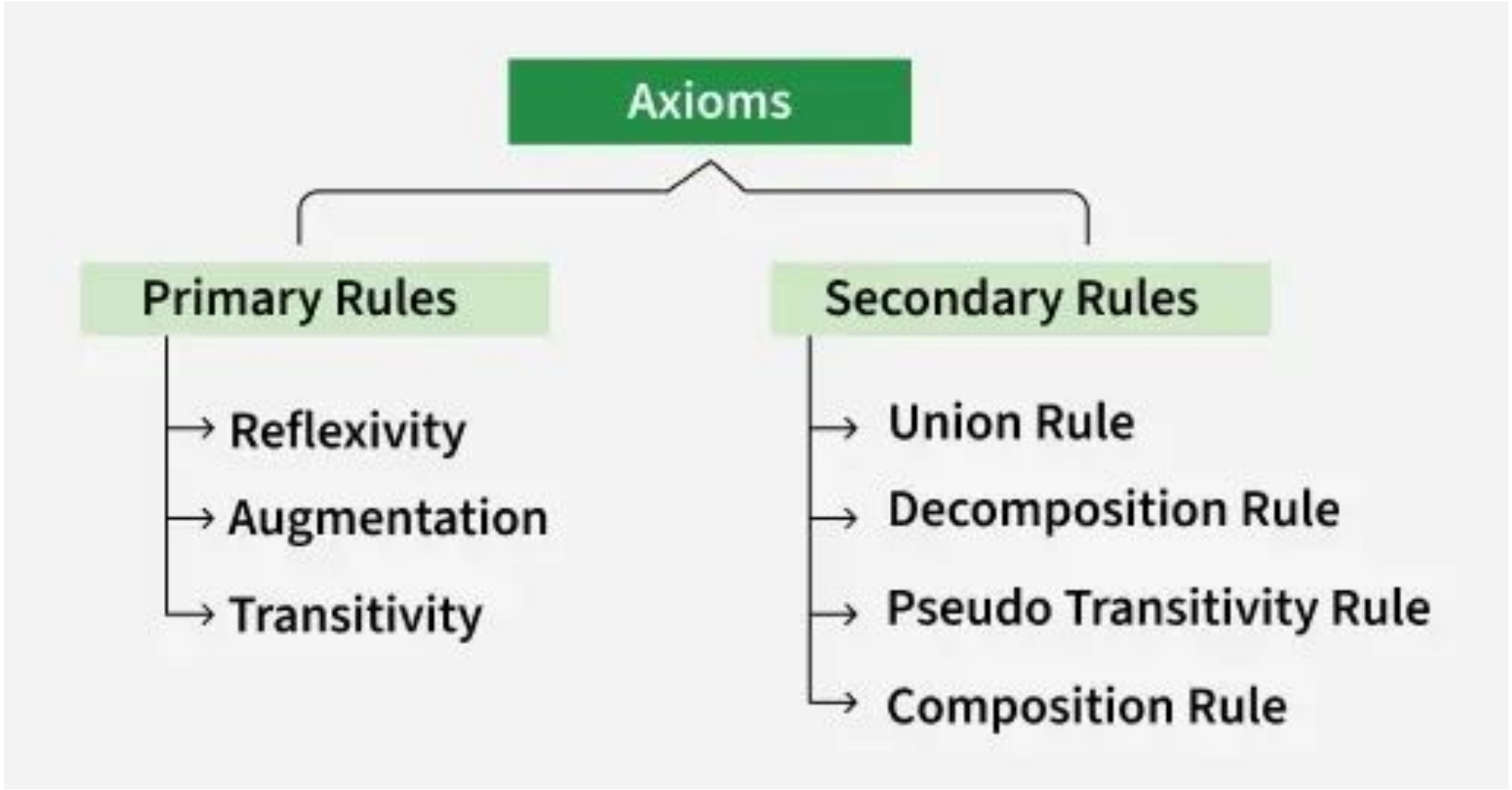
Armstrong Axioms

Armstrong's Axioms in Functional Dependency in DBMS

- Armstrong's Axioms refer to a **set of inference rules** that are **used to test the logical implication of functional dependencies**.
- Given a set of functional dependencies F, the closure of F (denoted as F+) is the set of all functional dependencies logically implied by F.
- Armstrong's Axioms, when applied repeatedly, help generate the closure of functional dependencies.
- These axioms are fundamental in determining functional dependencies in databases and are used to derive conclusions about the relationships between attributes.


Axioms
Primary Rules
Reflexivity
Augmentation
Transitivity
Secondary Rules
Union Rule
Decomposition Rule
Pseudo Transitivity Rule
Composition Rule

Primary Rules:

Axiom of Reflexivity:

If A is a set of attributes and B is a subset of A, then A holds B. If $B \subseteq A$ then $A \rightarrow B$.

This property is trivial property.

Axiom of Augmentation:

If **A**→**B** holds and Y is the attribute set, then **AY**→**BY** also holds.

That is adding attributes to dependencies, does not change the basic dependencies. If **A**→**B**, then **AC**→**BC** for any C.

Axiom of Transitivity:

Same as the transitive rule in algebra, if **A**→**B** holds and **B**→**C** holds, then **A**→**C** also holds.

In the dependency **A**→**B** we say that **A** functionally determines **B**. For example, if **X**→**Y** and **Y**→**Z**, then **X**→**Z.**

Secondary Rules:

Union:

If **A**→**B** holds and **A**→**C** holds, then **A**→**BC** holds. If **X**→**Y** and **X**→**Z** then **X**→**YZ**.

Composition:

If **A**→**B** and **X**→**Y** hold, then **AX**→**BY** holds.

Decomposition:

If **A**→**BC** holds then **A**→**B** and **A**→**C** hold. If **X**→**YZ** then **X**→**Y** and **X**→**Z**.

Pseudo Transitivity:

If **A**→**B** holds and **BC**→**D** holds, then **AC**→**D** holds. If **X**→**Y** and **YZ**→**W** then **XZ**→**W**.